12.471

EE25BTECH11042 - Nipun Dasari

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Question

The solution to system of equations

$$\begin{pmatrix} 2 & 5 \\ -4 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ -30 \end{pmatrix} \tag{0.1}$$

is

Theoretical Solution

We can write the given system in the form of an augmented matrix and solve using row reduction.

$$\left(\begin{array}{cc|c} 2 & 5 & 2 \\ -4 & 3 & -30 \end{array}\right) \tag{0.2}$$

Applying the row operation $R_2 \to R_2 + 2R_1$:

$$\left(\begin{array}{cc|c} 2 & 5 & 2 \\ 0 & 13 & -26 \end{array}\right) \tag{0.3}$$

Theoretical Solution

Applying the row operation $R_2 \to \frac{1}{13}R_2$:

$$\left(\begin{array}{cc|c} 2 & 5 & 2 \\ 0 & 1 & -2 \end{array}\right) \tag{0.4}$$

Applying the row operation $R_1 \to R_1 - 5R_2$:

$$\left(\begin{array}{cc|c} 2 & 0 & 12 \\ 0 & 1 & -2 \end{array}\right) \tag{0.5}$$

Theoretical Solution

Applying the row operation $R_1 \to \frac{1}{2}R_1$:

$$\left(\begin{array}{cc|c} 1 & 0 & 6 \\ 0 & 1 & -2 \end{array}\right) \tag{0.6}$$

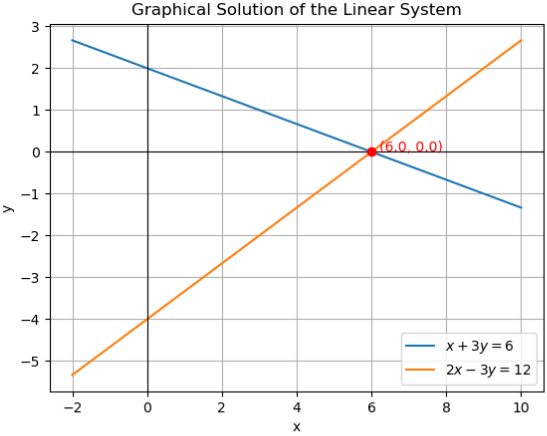
From the reduced row echelon form, we get the solution:

$$x = 6 \tag{0.7}$$
$$y = -2 \tag{0.8}$$

The solution vector is

$$\vec{x} = \begin{pmatrix} 6 \\ -2 \end{pmatrix} \tag{0.9}$$

Plot



Plot of the intersecting lines